

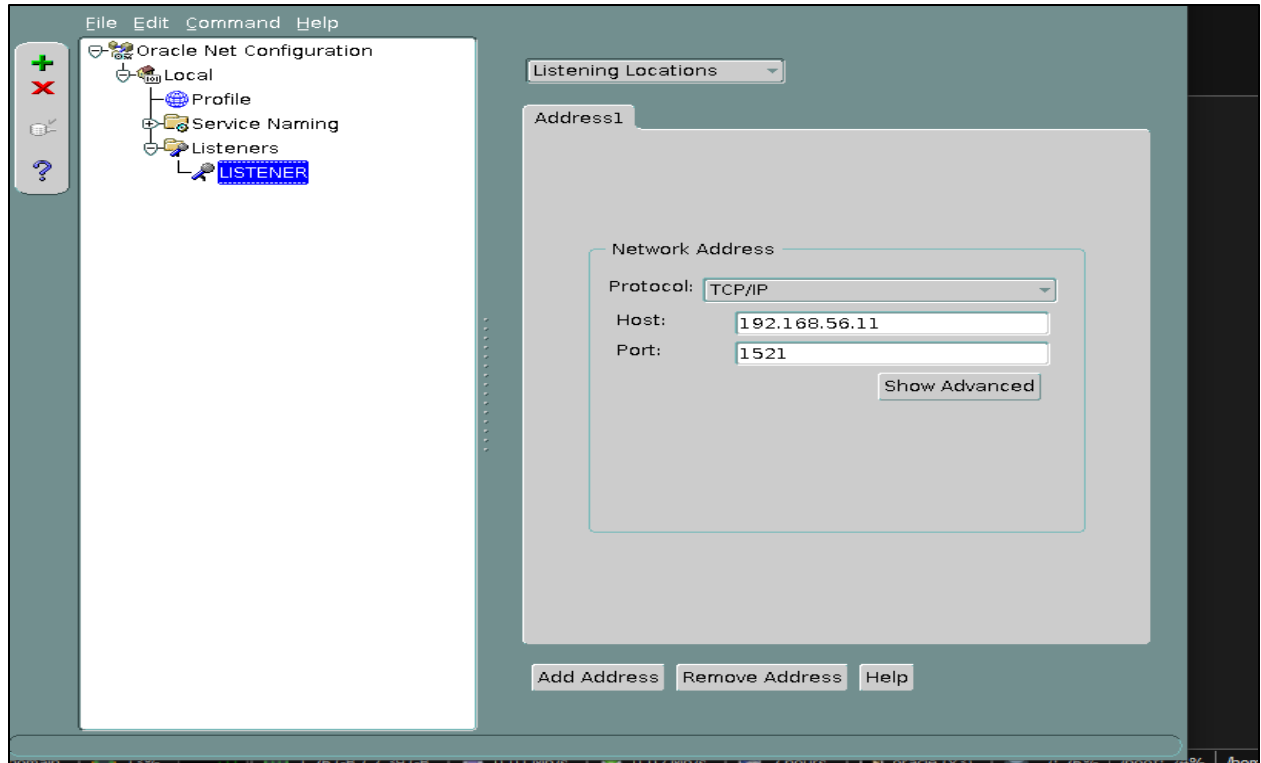
LISTENER & TNS

- Create listener with port 1721 via netmgr.

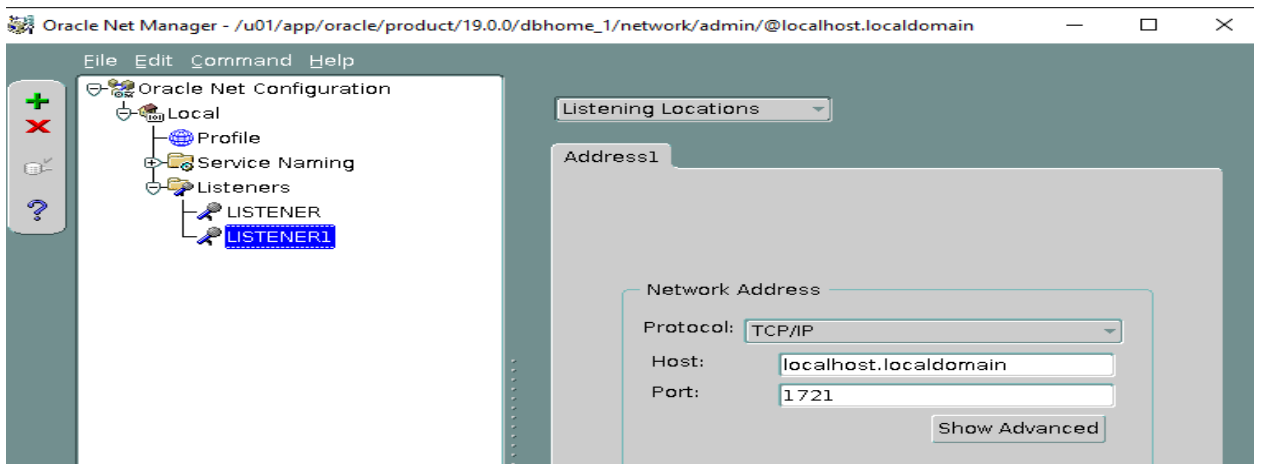
STEP 1: Check the listener and TNS location

```
[oracle@localhost ~]$ cd $ORACLE_HOME/network/admin
[oracle@localhost admin]$ pwd
/u01/app/oracle/product/19.0.0/dbhome_1/network/admin
```

STEP 2: Enter the Listener configuration with “NETMGR” or “NETCA” tools.



STEP 3: Now configure the listener



STEP 4: Start listener

```
[oracle@localhost admin]$ lsnrctl start listener1

LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 29-MAY-2026 20:36:38

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Starting /u01/app/oracle/product/19.0.0/dbhome_1/bin/tnslsnr: please wait...

TNSLSNR for Linux: Version 19.0.0.0.0 - Production
System parameter file is /u01/app/oracle/product/19.0.0/dbhome_1/network/admin/listener.ora
Log messages written to /u01/app/oracle/diag/tnslsnr/localhost/listener1/alert/log.xml
Listening on: (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=localhost)(PORT=1721)))

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=localhost.localdomain)(PORT=1721)))
STATUS of the LISTENER
-----
Alias                     listener1
Version                   TNSLSNR for Linux: Version 19.0.0.0.0 - Production
Start Date                29-MAY-2026 20:36:38
Uptime                    0 days 0 hr. 0 min. 0 sec
Trace Level               off
Security                  ON: Local OS Authentication
SNMP                      OFF
Listener Parameter File   /u01/app/oracle/product/19.0.0/dbhome_1/network/admin/listener.ora
Listener Log File         /u01/app/oracle/diag/tnslsnr/localhost/listener1/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=localhost)(PORT=1721)))
Services Summary...
Service "orcl" has 1 instance(s).
  Instance "orcl", status UNKNOWN, has 1 handler(s) for this service...
The command completed successfully
```

STEP 5: Check the configuration under the listener.ora file

```
[oracle@localhost admin]$ cat listener.ora
# listener.ora Network Configuration File: /u01/app/oracle/product/19.0.0/dbhome_1/network/admin/listener.ora
# Generated by Oracle configuration tools.

LISTENER1 =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = localhost.localdomain)(PORT = 1721))
  )

SID_LIST_LISTENER =
  (SID_LIST =
    (SID_DESC =
      (GLOBAL_DBNAME = orcl)
      (ORACLE_HOME = /u01/app/oracle/product/19.0.0/dbhome_1)
      (SID_NAME = orcl)
    )
  )

SID_LIST_LISTENER1 =
  (SID_LIST =
    (SID_DESC =
      (GLOBAL_DBNAME = orcl)
      (ORACLE_HOME = /u01/app/oracle/product/19.0.0/dbhome_1)
      (SID_NAME = orcl)
    )
  )

ADR_BASE_LISTENER1 = /u01/app/oracle

LISTENER =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = 192.168.56.11)(PORT = 1521))
  )

ADR_BASE_LISTENER = /u01/app/oracle
```

➤ **Create TNSnames file with manual method.**

STEP 1: Go to Network Admin Directory

```
[oracle@localhost admin]$  
[oracle@localhost admin]$ cd $ORACLE_HOME/network/admin
```

STEP 2: Create/Edit tnsnames.ora

```
[oracle@localhost admin]$ vi tnsnames.ora  
[oracle@localhost admin]$ cat tnsnames.ora  
ORCL =  
  (DESCRIPTION =  
    (ADDRESS = (PROTOCOL = TCP)(HOST = localhost)(PORT = 1721))  
    (CONNECT_DATA =  
      (SERVER = DEDICATED)  
      (SERVICE_NAME = orcl)  
    )  
  )
```

STEP 3: Test TNS Connectivity

```
[oracle@localhost admin]$ tnsping orcl  
TNS Ping Utility for Linux: Version 19.0.0.0.0 - Production on 29-MAY-2026 20:47:32  
Copyright (c) 1997, 2019, Oracle. All rights reserved.  
Used parameter files:  
  
Used TNSNAMES adapter to resolve the alias  
Attempting to contact (DESCRIPTION = (ADDRESS = (PROTOCOL = TCP)(HOST = localhost)(PORT = 1721)) (CONNECT_DATA = (SERVER = DEDICATED) (SERVICE_NAME = orcl)))  
OK (0 msec)
```

➤ **Perform DB connectivity using service name:**

-> **When Listener is down.**

-> **When Listener is up.**

DB Connectivity When Listener is UP

STEP 1: Check Listener Status.

```
[oracle@localhost admin]$ lsnrctl status listener1  
LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 29-MAY-2026 20:52:15  
Copyright (c) 1991, 2019, Oracle. All rights reserved.  
  
Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=localhost.localdomain)(PORT=1721)))  
STATUS of the LISTENER  
-----  
Alias                listener1  
Version              TNSLSNR for Linux: Version 19.0.0.0.0 - Production  
Start Date           29-MAY-2026 20:52:02  
Uptime               0 days 0 hr. 0 min. 12 sec  
Trace Level          off  
Security             ON: Local OS Authentication  
SNMP                 OFF  
Listener Parameter File /u01/app/oracle/product/19.0.0/dbhome_1/network/admin/listener.ora  
Listener Log File    /u01/app/oracle/diag/tnslsnr/localhost/listener1/alert/log.xml  
Listening Endpoints Summary...  
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=localhost)(PORT=1721)))  
Services Summary...  
Service "orcl" has 1 instance(s).  
  Instance "orcl", status UNKNOWN, has 1 handler(s) for this service...  
The command completed successfully
```

STEP 2: Connect Using Service Name.

```
[oracle@localhost admin]$ tnsping orcl

TNS Ping Utility for Linux: Version 19.0.0.0.0 - Production on 29-MAY-2026 20:53:01

Copyright (c) 1997, 2019, Oracle. All rights reserved.

Used parameter files:

Used TNSNAMES adapter to resolve the alias
Attempting to contact (DESCRIPTION = (ADDRESS = (PROTOCOL = TCP)(HOST = localhost)(PORT = 1721)) (CONNECT_DATA = (SERVER = DEDICATED) (SERVICE_NAME = orcl)))
OK (10 msec)
```

```
[oracle@localhost admin]$ sqlplus scott/tiger@orcl

SQL*Plus: Release 19.0.0.0.0 - Production on Fri May 29 20:53:22 2026
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Last Successful login time: Fri May 29 2026 20:12:46 +05:30

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> show user
USER is "SCOTT"
```

Connection successful because:

1. Listener is running
2. Service is registered with listener

DB Connectivity When Listener is DOWN

STEP 1: Stop Listener.

```
[oracle@localhost admin]$
[oracle@localhost admin]$ lsnrctl stop listener1

LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 29-MAY-2026 20:57:28

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=localhost.localdomain)(PORT=1721)))
The command completed successfully
[oracle@localhost admin]$
[oracle@localhost admin]$ sqlplus scott/tiger@orcl

SQL*Plus: Release 19.0.0.0.0 - Production on Fri May 29 20:57:38 2026
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

ERROR:
ORA-12541: TNS:no listener

Enter user-name: █
```

Connection interrupted Because:

1. Service name connection requires listener
2. Listener is not available

➤ **How to stop, start the Listener Process?**

1. Start Listener => lsnrctl start listener
2. Status Listener => lsnrctl status listener
3. Stop Listener => lsnrctl stop listener

➤ **How to check status of listener?**

```
[oracle@localhost admin]$ lsnrctl

LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 29-MAY-2026 21:04:23

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Welcome to LSNRCTL, type "help" for information.

LSNRCTL> status listener1
Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=localhost.localdomain)(PORT=1721)))
STATUS of the LISTENER
-----
Alias                     listener1
Version                   TNSLSNR for Linux: Version 19.0.0.0.0 - Production
Start Date                29-MAY-2026 21:02:58
Uptime                    0 days 0 hr. 1 min. 30 sec
Trace Level               off
Security                  ON: Local OS Authentication
SNMP                      OFF
Listener Parameter File   /u01/app/oracle/product/19.0.0/dbhome_1/network/admin/listener.ora
Listener Log File         /u01/app/oracle/diag/tnslsnr/localhost/listener1/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=localhost)(PORT=1721)))
Services Summary...
Service "orcl" has 1 instance(s).
  Instance "orcl", status UNKNOWN, has 1 handler(s) for this service...
The command completed successfully
```

➤ **How to check listener status from OS prompt?**

```
[oracle@localhost admin]$ lsnrctl status listener1

LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 29-MAY-2026 21:03:06

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=localhost.localdomain)(PORT=1721)))
STATUS of the LISTENER
-----
Alias                     listener1
Version                   TNSLSNR for Linux: Version 19.0.0.0.0 - Production
Start Date                29-MAY-2026 21:02:58
Uptime                    0 days 0 hr. 0 min. 7 sec
Trace Level               off
Security                  ON: Local OS Authentication
SNMP                      OFF
Listener Parameter File   /u01/app/oracle/product/19.0.0/dbhome_1/network/admin/listener.ora
Listener Log File         /u01/app/oracle/diag/tnslsnr/localhost/listener1/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=localhost)(PORT=1721)))
Services Summary...
Service "orcl" has 1 instance(s).
  Instance "orcl", status UNKNOWN, has 1 handler(s) for this service...
The command completed successfully
```

➤ How to kill listener process(PID) from OS prompt?

```
[oracle@localhost admin]$  
[oracle@localhost admin]$ ps -ef|grep tnslnsr  
oracle 10573 1 0 21:02 ? 00:00:00 /u01/app/oracle/product/19.0.0/dbhome_1/bin/tnslnsr listener1 -inherit  
oracle 11708 10457 0 21:05 pts/1 00:00:00 grep --color=auto tnslnsr  
[oracle@localhost admin]$  
[oracle@localhost admin]$  
[oracle@localhost admin]$ kill -9 10573  
[oracle@localhost admin]$ ps -ef|grep tnslnsr  
oracle 12206 10457 0 21:06 pts/1 00:00:00 grep --color=auto tnslnsr  
[oracle@localhost admin]$
```

➤ What command you will use to reload listener?

```
[oracle@localhost ~]$ lsnrctl reload listener1  
LSNRCTL for Linux: Version 19.0.0.0 - Production on 29-MAY-2026 21:36:09  
Copyright (c) 1991, 2019, Oracle. All rights reserved.  
Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=localhost.localdomain)(PORT=1721)))  
The command completed successfully
```

➤ How to check listener logs?

```
[oracle@localhost trace]$ pwd  
/u01/app/oracle/diag/tnslnsr/localhost/listener1/trace  
[oracle@localhost trace]$  
[oracle@localhost trace]$  
[oracle@localhost trace]$  
[oracle@localhost trace]$ ls  
listener1.log
```

```
[oracle@localhost trace]$ ls  
listener1.log  
[oracle@localhost trace]$ tail -f listener1.log  
TIMESTAMP * CONNECT DATA [* PROTOCOL INFO] * EVENT [* SID] * RETURN CODE  
WARNING: Subscription for node down event still pending  
29-MAY-2026 21:35:59 * (CONNECT_DATA=(CID=(PROGRAM=)(HOST=localhost.localdomain)(USER=oracle))(COMMAND=status)(ARGUMENTS=64)(SERVICE=listener1  
) (VERSION=318767104)) * status * 0  
2026-05-29T21:36:09.969613+05:30  
System parameter file is /u01/app/oracle/product/19.0.0/dbhome_1/network/admin/listener.ora  
Trace information written to /u01/app/oracle/diag/tnslnsr/localhost/listener1/trace/ora_13353_139760273914944.trc  
Trace level is currently 0  
Log messages written to /u01/app/oracle/diag/tnslnsr/localhost/listener1/alert/log.xml  
29-MAY-2026 21:36:09 * (CONNECT_DATA=(CID=(PROGRAM=)(HOST=localhost.localdomain)(USER=oracle))(COMMAND=reload)(ARGUMENTS=64)(SERVICE=listener1  
) (VERSION=318767104)) * reload * 0  
^Z  
[1]+ Stopped tail -f listener1.log  
[oracle@localhost trace]$  
[oracle@localhost trace]$  
[oracle@localhost trace]$ tail -4f listener1.log  
Trace information written to /u01/app/oracle/diag/tnslnsr/localhost/listener1/trace/ora_13353_139760273914944.trc  
Trace level is currently 0  
Log messages written to /u01/app/oracle/diag/tnslnsr/localhost/listener1/alert/log.xml  
29-MAY-2026 21:36:09 * (CONNECT_DATA=(CID=(PROGRAM=)(HOST=localhost.localdomain)(USER=oracle))(COMMAND=reload)(ARGUMENTS=64)(SERVICE=listener1  
) (VERSION=318767104)) * reload * 0
```

➤ List out some ORA error if user is not able to do connectivity using service name due to listener issues?

Listener issues can cause connectivity errors such as ORA-12541, ORA-12514, ORA-12154, and ORA-12505 during service name connection attempts.